📘 Chapter 1: Why Do We Model Things? (25 MCQs)

1. What defines the boundary of a system?
   a) Budget
   b) Components' limits
   c) Objectives
   d) External actors
2. The components of a system have:
   a) General roles
   b) Random functionality
   c) Specific responsibilities
   d) Equal power distribution
3. What should the communication between system components be like?
   a) Continuous
   b) As frequent as possible
   c) Only on a need basis
   d) Through external tools
4. What is a model in software engineering?
   a) A working software
   b) A simulation tool
   c) A simplification of reality
   d) A database
5. Why do we build models of complex systems?
   a) To avoid testing
   b) Because models are expensive
   c) Because we can’t understand the entire system at once
   d) Because they replace documentation
6. Modeling helps reduce complexity by:
   a) Eliminating all components
   b) Visualizing only errors
   c) Focusing on one aspect at a time
   d) Ignoring user input
7. What is a key feature of modeling in software engineering?
   a) Final implementation
   b) Designing before coding
   c) Enhancing internet speed
   d) Reducing hardware usage
8. Which of the following is NOT a reason for modeling?
   a) Helps us understand systems
   b) Replaces testing
   c) Essential for large projects
   d) Proven engineering technique
9. Why do successful projects use modeling?
   a) Because it’s legally required

b) To delay implementation
c) To better understand the system
d) To outsource work

10. What is NOT a benefit of using a model?
    a) Visualization of a system
    b) Replacement of developers
    c) Specification of system behavior
    d) Documentation of decisions
11. Models help us to:
    a) Reduce performance
    b) Visualize, specify, construct, and document
    c) Eliminate design
    d) Minimize project scope
12. What kind of decisions can models help document?
    a) Business names
    b) Team salaries
    c) Development choices
    d) Deployment costs
13. The right model will:
    a) Confuse developers
    b) Slow down design
    c) Help solve problems
    d) Replace UML
14. Every model may be expressed:
    a) Once
    b) At different levels of precision
    c) Only at runtime
    d) As a table
15. Which of the following is a good characteristic of models?
    a) Unrelated to reality
    b) Unclear and abstract
    c) Connected to reality
    d) Too detailed
16. What happens if a model is too abstract?
    a) Becomes easier
    b) Needs to be closer to reality
    c) Is perfect
    d) Will auto-code itself
17. What is true about system modeling?
    a) One model is enough
    b) Models should be completely disconnected
    c) Multiple models are better
    d) Only one view matters
18. Models should be chosen based on:
    a) Aesthetic design
    b) Convenience

c) How they influence solution shaping
d) How colorful they are

19. What makes a model helpful in development?
    a) It’s long and complicated
    b) It shows every line of code
    c) It helps visualize and plan
    d) It removes team communication
20. What is the role of modeling in small and medium projects?
    a) Irrelevant
    b) Not needed
    c) Helpful
    d) Mandatory only in Agile
21. The output of one component becomes:
    a) An isolated file
    b) Another system
    c) Input for another component
    d) A test case
22. In modeling, specifying structure or behavior refers to:
    a) Visualization only
    b) Implementation
    c) Modeling detail
    d) Code translation
23. Models act as a:
    a) Replacement for engineers
    b) Magic wand
    c) Blueprint for system construction
    d) Test report
24. A system should have objectives that are:
    a) Broad and general
    b) Open to interpretation
    c) Undefined
    d) Specific and well-defined
25. Components should communicate:
    a) Only during emergencies
    b) As much as possible
    c) Minimally and purposefully
    d) Through email

📗 Chapter 2: Introduction to UML (25 MCQs)

1. UML stands for:
    a) Uniform Modeling Logic
    b) Unified Method Language

c) Unified Modeling Language
d) Universal Mapping Layout

2. UML was created and is maintained by:
   a) W3C
   b) Microsoft
   c) Object Management Group
   d) IEEE
3. UML is primarily used for:
   a) Testing hardware
   b) Visualizing and modeling software systems
   c) Web design
   d) Operating system installation
4. Which of the following is NOT a function of UML?
   a) Visualization
   b) Specification
   c) Construction
   d) Compiling
5. UML can be interpreted by:
   a) End users
   b) Programming tools and developers
   c) Antivirus programs
   d) Internet browsers
6. What kind of models does UML support?
   a) Only static models
   b) Only flowcharts
   c) Static and dynamic models
   d) Voice-based models
7. UML models can be mapped to:
   a) Paint drawings
   b) RDBMS tables or programming languages
   c) UML diagrams only
   d) Spreadsheets
8. One purpose of using UML is:
   a) Hiding design flaws
   b) Avoiding team collaboration
   c) Documenting system architecture
   d) Skipping project planning
9. The vocabulary of UML includes:
   a) Code snippets
   b) Only diagrams
   c) Things, relationships, diagrams
   d) Only interfaces
10. Structural things in UML are usually:
    a) Verbs
    b) Relationships

c) Nouns
d) Packages

11. Behavioral things in UML represent:
    a) Configuration files
    b) Verbs or dynamic behavior
    c) Static views
    d) System logs
12. A class in UML is defined as:
    a) An executable file
    b) A blueprint for objects
    c) A folder
    d) A static diagram
13. Which of the following can NOT be instantiated?
    a) A class
    b) An object
    c) An interface
    d) A component
14. What is a collaboration in UML?
    a) A use case
    b) A data type
    c) A group of UML elements working together
    d) An interface
15. A use case in UML is used to describe:
    a) Class behavior
    b) Sequence of actions with a result
    c) Code compilation
    d) Loops in code
16. An active class contains:
    a) Comments
    b) Threads/processes
    c) Only data
    d) Flowcharts
17. A component in UML represents:
    a) Code errors
    b) An HTML page
    c) A module/package of classes
    d) A single object
18. A node in UML is:
    a) A static class
    b) A comment
    c) A hardware resource
    d) An object method
19. Annotations in UML are:
    a) System logs
    b) Code snippets

c) Notes or comments
d) Input fields

20. Which is NOT a UML relationship?
    a) Association
    b) Aggregation
    c) Realization
    d) Dependency
21. Generalization represents:
    a) A “has-a” relationship
    b) A data structure
    c) An inheritance (IS-A) relationship
    d) A comment type
22. Realization is the implementation of:
    a) A class
    b) A flowchart
    c) An interface
    d) A data structure
23. Which diagram shows time-ordered interactions?
    a) Class diagram
    b) Sequence diagram
    c) Deployment diagram
    d) Object diagram
24. A use case view exposes:
    a) Deployment layout
    b) System performance
    c) System behavior
    d) Compilation time
25. In RUP, what comes after “Elaboration” phase?
    a) Transition
    b) Planning
    c) Construction
    d) Maintenance